

⚡ GENAI INFRASTRUCTURE & LLMOPS PREP

15+

Memory, Databases & Caching for GenAI

INTERVIEW QUESTIONS

📖 WHAT YOU'LL MASTER INSIDE

Every Answer Explained Simply

Short & Long-Term Memory • Relational vs Vector DBs • Document Ingestion • HNSW & Hybrid Search • Multi-Tenant Isolation

Q1. How would you design short-term and long-term memory for an AI agent performing multi-step tasks?

Short-term memory holds working context in the LLM context window; long-term memory stores durable facts across sessions in external databases.

Memory Type	Short-Term Memory	Long-Term Memory
Storage Layer	Active Context Window (RAM)	Vector + Relational DBs
Primary Content	Task plan, tool outputs, steps	User facts, past decisions, domain knowledge
Key Constraint	Strict token limit (e.g. 128k)	Retrieval latency & relevance
Management	Token counting & step compression	Semantic search & recency decay

🔄 THE MEMORY GRADUATION HANDOFF

When a task concludes, durable findings (e.g., user preferences, verified workflow solutions) **graduate** from working context into long-term vector/relational tables so future sessions retrieve them automatically.

🔖 Check caption for complete AI Engineer Interview Kit.

Q2. What architectural challenges arise when storing conversation history for long-running chat sessions?

Long-running sessions create compounding engineering hurdles around storage throughput, token window saturation, and state tracking.

- 1 Storage Throughput at Scale:** Millions of active turns require relational indexing (`user_id`, `session_id`, `turn_number`) optimized for high write concurrency and slice reads.
- 2 Context Injection Strategy:** You cannot load 300 turns. Combine **Recency** (last 5-10 turns verbatim) + **Semantic Search** (top-k relevant turns) + **Anchor turns** (first 2 turns).
- 3 Structured Session State:** Beyond chat text, store decisions and preferences in structured JSON records injected as dense summaries rather than raw dialogue logs.

✓ BEST PRACTICE ARCHITECTURE

Never pass raw full history. Store immutable turns in SQL, index embeddings in a vector store, and assemble context dynamically per turn.

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Q3. How would you prevent conversation memory from exceeding the model's context window?

Implement proactive token budgeting and a hierarchical compression strategy before hitting the prompt context limit.

```
def assemble_context(system, history, query, max_tokens=100000):  
    budget = max_tokens - 2000 # reserve output buffer  
    reserved = count_tokens(system) + count_tokens(query)  
    return fit_history_to_budget(history, budget - reserved)
```

📌 3-TIER COMPRESSION HIERARCHY

- **Tier 1: Tool Output Summarization:** Replace 500-token JSON tool returns with 20-token key findings once consumed.
- **Tier 2: Rolling Summarization:** Every 20 turns, run a background LLM job to collapse older blocks into a 1-paragraph summary.
- **Tier 3: Truncation Gate:** Retain only the rolling summary + last N turns + top-k semantically matched turns.

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Q4. What strategies allow an AI system to remember important facts while discarding irrelevant context?

Distinguish high-value signal from conversational chatter using explicit extraction, decay functions, and async consolidation.

```
// Structured Fact Extraction Output
{
  "user_preferences": ["prefers bullet points", "fintech industry"],
  "established_facts": ["project launch: March 15"],
  "user_decisions": ["selected GPT-4o for production"]
}
```

- 1 Explicit Importance Signals:** Extract statements with phrases like "I always prefer" or "remember that" directly into structured JSON tables.
- 2 Decay-Based Forgetting:** Weight vector retrieval scores by Semantic Similarity $\times$ Recency Decay so unaccessed older notes fade.
- 3 Async Consolidation:** Periodically deduplicate memory; mark old preferences as "superseded" when users change directions.



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Q5. How would you design memory systems for multi-user AI applications where sessions must remain isolated?

Multi-tenant isolation requires mandatory hard filtering at the database access layer —never rely solely on LLM prompt instructions.

```
def get_user_memories(user_id: str, query_emb: list) -> list:
    return vector_store.search(
        embedding=query_emb,
        mandatory_filters={"user_id": user_id}, # infrastructure layer
        top_k=10
    )
```

DEFENSE-IN-DEPTH ISOLATION STACK

-  **Tenant Scoped Schema:** Every database row & vector metadata payload must contain indexed `user_id` and `tenant_id` keys.
-  **Redis Cache Namespacing:** Namespace keys strictly as `session:{user_id}:{session_id}:state` to prevent key collisions.
-  **System Prompt Scoping:** System prompts instruct the LLM to assist only the current user ID as an added guardrail.

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2. DATABASES FOR GENAI

Q6. When designing a GenAI system, how do you decide between relational, document, and vector databases?

Modern GenAI systems use a polyglot persistence architecture combining structured SQL with semantic vector indexes.

DB Type	Optimal AI Workload	Key Technologies
Relational	ACID users, billing, audit logs, metadata filters	PostgreSQL (pgvector)
Document	Schema-flexible agent execution traces, logs	MongoDB, DynamoDB
Vector DB	Large-scale approximate nearest neighbor search	Pinecone, Qdrant, Milvus



PRODUCTION POLYGLOT STACK

In practice, **PostgreSQL + pgvector** handles up to 10M vectors alongside structured metadata. Dedicated vector databases scale when datasets exceed tens of millions of embeddings.



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2. DATABASES FOR GENAI

Q7. What advantages do vector databases provide compared to traditional databases in AI applications?

Purpose-built vector infrastructure delivers specialized approximate search algorithms and distributed embedding management.

⚡ OPTIMIZED ANN SEARCH

Hardware-tuned HNSW, DiskANN, and IVF indexes achieve sub-50ms latency across 100M+ vectors.

🔍 SINGLE-STAGE HYBRID

Execute combined semantic vector similarity + structured metadata pre-filtering in one pipeline.

⚙️ LIFECYCLE MANAGEMENT

Built-in sharding, replication, dynamic quantization, and incremental background graph updates.

🔗 HORIZONTAL SCALE

Distributed graph partitioning designed specifically for high-dimensional floating-point vectors.



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Q8. How would you structure a database schema to support document ingestion for RAG pipelines?

Separate parent document metadata from chunk embeddings to support atomic invalidation, deduplication, and ACL filters.

```
CREATE TABLE documents (id UUID PRIMARY KEY, file_hash VARCHAR(64),
status VARCHAR(20), version INT);
CREATE TABLE document_chunks (
    id UUID PRIMARY KEY, document_id UUID REFERENCES documents(id),
    chunk_index INT, embedding_model VARCHAR(100), embedding_version
INT,
    vector_store_id VARCHAR(200)
);
```

WHY THIS SCHEMA WORKS

- ✓ **Cascade Invalidation:** Linking chunks to document_id lets you supersede all chunks atomically on document updates.
- ✓ **Embedding Model Tracking:** Tracking embedding_model allows incremental migrations when upgrading embedding dimensions.
- ✓ **ACL Governance:** Join against a document_permissions table at retrieval time to enforce user read access.

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Q9. What challenges arise when storing millions of embeddings in production systems?

At billion-scale, bottlenecks shift from computation to RAM capacity, index build latency, and disk IOPS trade-offs.

Index Strategy	Memory Footprint	Latency & Scale
In-Memory HNSW	High (Full graph in RAM)	Lowest latency (<10ms), costly at scale
DiskANN / SSD	Low (Graph stored on NVMe)	High capacity, slight IO overhead
IVF-PQ	Compressed (8x–32x smaller)	Scales to billions, minor recall drop

⚡ PRODUCT QUANTIZATION (PQ) ADVANTAGE

When full-precision float32 embeddings exceed RAM capacity, apply **Product Quantization (IVF-PQ)** or scalar quantization (int8) to compress index footprint by up to 4x–8x with negligible recall degradation.



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2. DATABASES FOR GENAI

Q10. How would you update embeddings when source documents change frequently?

Implement an atomic shadow-ingestion workflow to prevent stale RAG retrievals or downtime during index synchronization.

1

Priority Webhook Queue: Listen to source change APIs (Confluence/Drive). Enqueue updates prioritized by document query traffic.

2

Shadow Ingestion: Parse, chunk, and embed the updated version into the vector store marked with status active.

3

Atomic Swap & Supersede: Delete old vector records and update SQL chunks to status superseded in a single transaction window.

⚠ CORRECTNESS GUARDRAIL

Always filter vector queries on `status = active` so queries never return duplicate or half-deleted chunks during the update window.



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Q11. What indexing strategies help vector databases perform similarity search efficiently?

Exact k-NN search $O(N \times D)$ is too slow for real-time AI. ANN algorithms structure vector space for logarithmic navigation.

ANN Index	Core Data Structure	Best Production Use Case
HNSW	Multi-layer navigable small world graphs	Standard RAG up to 10M-50M vectors (highest recall)
IVF	K-means inverted Voronoi cluster centroids	Fast indexing when memory is constrained
IVF-PQ	Inverted clusters + product quantization	100M+ billion-scale vector collections

⚙️ CRITICAL HNSW TUNING PARAMETERS

ef_construction: Index quality vs build time • **ef_search:** Query recall vs search latency • **M:** Graph edge density per node.

📖 Check caption for complete AI Engineer Interview Kit.

Q12. How would you design hybrid search combining vector similarity and keyword search?

Dense embeddings capture semantic meaning; sparse keyword search (BM25) captures exact SKU codes, IDs, and names.

$$\text{RRF_Score} = \sum 1 / (60 + \text{rank_i})$$

HYBRID SEARCH PIPELINE

- ⚡ **Parallel Retrieval:** Execute dense vector search and sparse BM25 keyword search concurrently across candidate lists.
- ⚡ **Reciprocal Rank Fusion (RRF):** Merge ranked results without score normalization issues using inverse rank summation.
- ⚡ **Dynamic Query Weighting:** Boost BM25 weight automatically when exact SKU IDs or proper nouns are detected in the query.

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Q13. What architectural factors influence vector search latency at scale?

End-to-end RAG latency is governed by embedding inference, index memory residency, and payload fetch round-trips.

EMBEDDING API LATENCY

Query embedding generation (50–150ms) frequently dominates total RAG latency over ANN search.

INDEX PLACEMENT

In-memory HNSW responds in <10ms; disk-backed SSD search adds NVMe read latency.

PRE VS POST FILTERING

Metadata post-filtering ruins recall; single-stage HNSW pre-filtering maintains speed.

PAYLOAD HYDRATION

Store chunk text directly or batch-fetch from SQL to prevent N+1 document round-trips.

 Check caption for complete AI Engineer Interview Kit.

Q14. How would you evaluate retrieval quality in a vector database system?

Systematic golden dataset evaluation prevents RAG regressions when changing chunking, models, or index parameters.

Metric	What It Measures	Interpretation
Recall@K	Fraction of relevant docs in top-K	Measures if true evidence is retrieved at all
MRR	Mean Reciprocal Rank of 1st hit	Penalizes systems where correct doc is buried
NDCG@K	Ranked graded relevance score	Rewards having the best document at position 1

📈 ONLINE PRODUCTION TELEMETRY

Complement offline benchmarks by tracking user **query reformulations** and citation click-through rates. Immediate user rephrasing signals poor top-K retrieval.



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Q15. What failure scenarios must be considered when deploying vector search in production?

Vector search failures are frequently silent—degrading RAG accuracy without throwing hard service exceptions.

! 4 CRITICAL PRODUCTION FAILURE MODES

- ✖ **Embedding API Outage:** Implement circuit breakers that fall back to lexical BM25 search when embedding endpoints fail.
- ✖ **Silent Index Corruption:** Reconcile vector counts against SQL rows daily and maintain automated snapshot backups.
- ✖ **Memory Pressure Eviction:** Monitor OS RSS RAM; page-swapping index segments causes 10x latency spikes.
- ✖ **Cold Start Thundering Herd:** Enforce readiness probes so instances fully load HNSW graphs before serving traffic.

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GenAI Memory & Retrieval Decision Matrix

Use this production roadmap to select optimal persistence and indexing architectures for LLM systems.

Problem Requirement	Optimal Technical Solution
Multi-step Working Context	Active Context Window + Step Compression
Long-Running Chat Sessions	SQL History + Recency + Semantic Retrieval
Multi-Tenant Data Privacy	Mandatory Infrastructure Filters (user_id)
SKUs + Semantic RAG Search	Hybrid Search (Dense + BM25) with RRF
Billion-Scale Vector Scale	IVF-PQ / DiskANN + SSD-backed Search

GOLDEN RULES FOR INTERVIEW SUCCESS

- ✓ Never pass raw full chat history into LLM context windows.
- ✓ Always mention Reciprocal Rank Fusion (RRF) for hybrid search.
- ✓ Highlight mandatory tenant ID filters at the DB layer for security.

 Check caption for complete AI Engineer Interview Kit.